Permanent Magnet Synchronous Motor for Electric Vehicle Applications

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***ABSTRACT*—Permanent Magnet Synchronous Motors have the potential to providing hign torque-to-current ratio, high power-to-weight ratio, high efficiency and robustness.Due to the above favorable points,PMSMs are commonly used in latest variable speed AC drives, particularly in Electric Vehicle applications. Electric vehicle is a nightmare for the city traffic without providing harmful gases with low noise. Permanent MagnetSynchronous Motor became at the top of ac motors in high performance drive systems such as EV. This paper proposed a radial flux inner rotor PMSM design for electric vehicle application. The PMSM design is The design is developed in Ansys Electronicsof the motor.**

Keywords—Electric vehicle; Radial flux; Ansys Maxwe;Permanent Magnet Synchronous Motor

I. INTRODUCTION

The dependence of fossil fuels has been step by step growing within the beyond century notwithstanding the famous fact that the resources of suchfuels are restrained. It is now widely familiar that one of these demand growth is not always sustainable. Stimulated not only by the destiny supply concerns however also by way of the concerns of overexploitation of fossil resources the hobby in electric powered and hybrid automobile programs has been developing rabidly in recent years. Low energy consumption and high operating efficiency are provided by permanent magnets. The field excitation of PMSM is provided by permanent magnets. PMSM has sinusoidal back EMF. That is similar to BLDC motor. Electric stability, durability, and reliable performance are provided by PMSM. Low energy consumption and high operating efficiency are the main advantages of permanent magnet motors. The growth of the PMSM in market is increased by its unique characteristic of usage in all type of AC electric motors. PMSMs provide sturdiness in addition to electric balance and make sure dependable overall performance as PMSM improves the motor efficiency approximately by 10%, it increases the usage of PMSM in various applications. The fuel consumption of PMSM is very low, so it is used in green vehicles. The awareness of green and electric vehicle improves the PMSM. Permanent Magnet Synchronous Motor were an increasing number of utilized in electric vehicle applications owing their high efficiency, excessive power density, compact structure and rapid dynamic response. High power to volume ratio and high efficiency are achieved by permanent magnet motors.

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II. BLOCK DIAGRAM

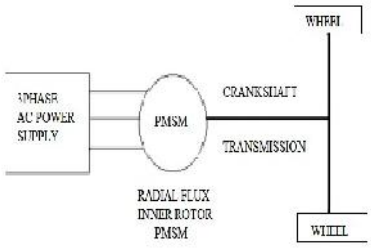

Fig. 2.1 proposed block diagram

The proposed block diagram shows in figure 1.1 contains 3phase AC power supply, radial flux surface mounted Permanent Magnet Synchronous Motor.

III. PERMANENT MAGNET SYNCHRONOUS MOTOR

The PMSM is an AC synchronous machine. The field excitation of PMSM is provided by the permanent magnet. PMSM is the go among Induction Motor and Brushless DC Motor with permanent magnet rotor and winding on stator. PMSM is cross between induction motor and Brushless DC Motor. The permanent magnet rotor and windings on the stator are similarly as the brushless DC motor.However, the stator shape with winding constructed is providing a sinusoidal flux density in the airgap of the motor as that of an Induction Motor.

3.1 CONSTRUCTION OF PMSM

It is same as the construction synchronous motor. The only difference is in the rotor has permanent magnet which is absent in the synchronous motor.The stator has winding which is excited by the electric power and the rotor is excited by its own permanent magnet. It is classified into the following two types

1. Surface Mounted PMSMs
2. Buried or interior PMSMs

In this project Interior or Buried PMSM is used shown in fig 4.1, the permanent magnets are embedded into the rotor instead of mounting on the surface of the rotor. This provides robustness and hence can be used in high speed applications. Reluctance torque is present in the interior PMSM because of the presence of saliency.

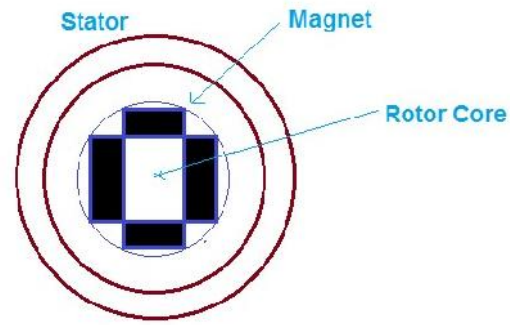

Fig. 3.1 Interior PMSM

3.2 WORKING PRINCIPLE OF PMSM

The working principle of Permanent Magnet Synchronous Motor is thatwhen the three phase winding of stator is energized by the power supply and the rotating magnetic field is generated in the airgap. The rotor poles lock the rotating magnetic field for producing the torque to rotate the shaft.

As we realize that synchronous motors are not self-starting, PMSM needs to be commenced one way or the other. Since there may be no windings at the rotor, induction windings for beginning are not relevant for such vehicles and therefore frequency strength deliver for this reason.

3.3 PMSM FOR ELECTRIC VEHICLE

As a motor for driving hybrid automobiles, the everlasting magnet (PM) motor is most customarily used. The reasons for this are: the magnetic field contemporary important for the induction system is unnecessary, that means secondary copper loss does not arise and the

motor is distinctly green; and the capability to elevate the magnetic flux density through uncommon earth magnets is a bonus for making the motor smaller.

IV. DESIGN OF PMSM USING RMxprt

Designers of electrical machines and mills can beautify ANSYS Maxwell with ANSYS RMxprt, a template-based totally layout tool. The following parameters are used in the Permanent Magnet Synchronous Motor.

- Stator
  - Slot
  - Winding
- Rotor
  - Pole

4.1.1 STATOR

It is the stationary part of the Permanent Magnet Synchronous Motor. Stator of the PMSM is distributed winding. The stacking issue is the ratio of electrical metallic along the axial period of the iron core. In the electric gadget layout software program Emetor, you may provide a stacking element for each the stator and rotor laminations (zero.1<=stacking factor<=1).

$$\frac{D^2 L}{t_m} \quad (1)$$

Where

Dis stator bore diameter

Lis stator core length

V is coefficient

Table 4.1 contains the following details of stator values.

Parameter	*Values*	*Units*
Outer diameter	500	mm
Inner diameter	410	mm
Length	120	mm
Stacking factor	0.95	
Steel type	M27_29G	
Number of slots	12	
Slot type	4	
Skew width	0	

Table. 5.1 Stator parameters

4.1.2 SLOT

The height and width of the slot in the stator of Permanent Magnet Synchronous Motor is given in the table 4.2.

Parameters	*Values*	*Units*
HS0	2	mm
HS1	3	Mm
HS2	17	Mm
BS0	11	Mm
BS1	235	Mm
BS2	29	Mm
RS	2	

Table. 5.2 Slot parameters

4.1.3 WINDING

Stator winding number of turns per phase can be calculated from the following equation.

$$V_{\varphi,r} = \sqrt{2}\pi f_e \dot{N}_a \emptyset_{g.p} \quad (2)$$

Where

$V_{\varphi,ed}$ is the rated phase voltage

f_e is the electrical frequency

$$N_a^{\cdot} = k_W N_a / 1.1 \quad (3)$$

$$\emptyset_p = \frac{2B_{g.p} D}{P} (4)$$

N_a is the number of series turns per phase

k_W is the winding factor

The relationship between Na and Nc of PMSM is shown in equation (5) and (6).

$$N_a = P_q N_c / C (5)$$

$$N_c = \frac{1.1 V_\varphi \quad C}{2\sqrt{2}\pi f_e q k_w B_{g.p} \quad D} (6)$$

The coefficient 1.1 inside the numerator is to atone for the leakage flux since it is not taken into consideration while deriving the equation.

The stator winding parameters of Permanent Magnet Synchronous Motor which is used in Electric Vehicle is given in the following table 4.3.

Parameters	*Values*
Winding layer	1
Winding type	Whole coiled
Parallel branch	1
Conductor per slot	4

Table 4.3 Stator winding parameter

4.1.3 ROTOR

The mechanical stress on the rotor is produced by the force, especially when the tip speed is very high. After the scale are decided, it is essential to check if the rotor diameter is inside the secure range. The outer diameter of rotor is calculated using the following equation.

$$D_{r,m} = \frac{v_c}{\omega_r} \quad (7)$$

Where

$D_{r,}$ is the maximum rotor diameter

v_c is critical speed depend on material property

is the angular velocity

The table 4.4 shows the parameters of rotor in the PMSM

Parameters	*Values*	*Units*
Outer diameter	210	Mm
Inner diameter	145	Mm
Length	75	Mm
Steel type	M27_27G	
Stacking factor	0.95	
Pole type	5	

Table 4.4 Rotor parameters

4.1.5 POLE

The magnetic pole used in PMSM is Vacodym744. The table 4.5 shows the parameter of pole in PMSM.

Parameters	*Values*	*Units*
Embrace	0.8	
Bridge	2	
Rit	0	
Magnetic material	Vacodym744	
Magnetic width	36	Mm
Thickness	6	Mm

Table 4.5 Pole parameters

4.2 ANALYSIS VALUES

The rated values assigned in PMSM are given in the table 4.6

Parameters	*Values*	*Units*
Fixed output power	3.5	Kw
Rated voltage	300	V
Rated speed	1500	RPM

Table 4.6 Pole parameters

V RESULT AND DISCUSSION

This chapter provides the detailed discussion about the obtained results. Simulation has been done to verify the output performance of the inner rotor permanent magnet synchronous motor. Figure 5.1 shows the design of permanent magnet synchronous motor by using Ansys Maxwell.

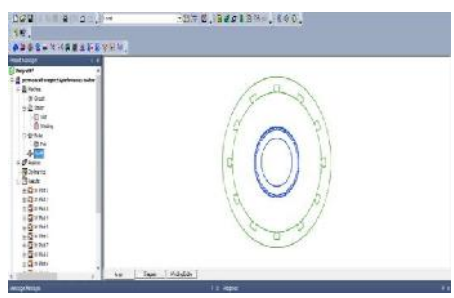

Fig 5.1. Design of PMSM using Ansys Maxwell

In the figure 5.2 input current value of PMSM is given

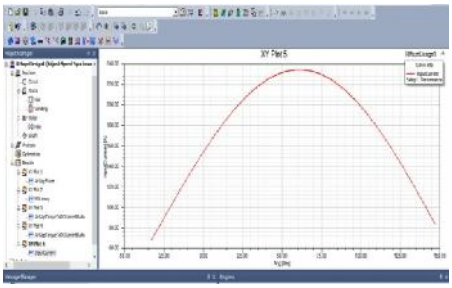

Fig 5.2 Input current value

Fig 5.3 shows the torque value of permanent magnet synchronous motor.

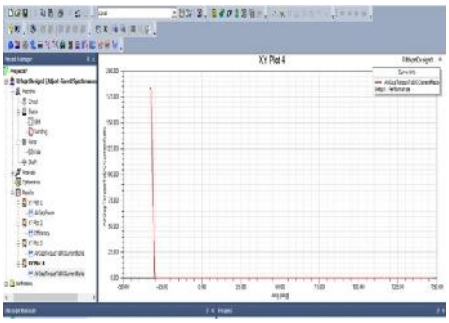

Fig 5.3 Air gap torque

Fig 5.4 shows the efficiency of the permanent magnet synchronous motor over 97% which is greater than the other electrical machines such as DC motor and induction motor.

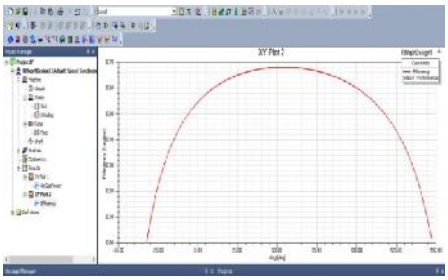

Fig 5.4 Efficiency of PMSM

In the figure 5.5, the airgap power curve is given.

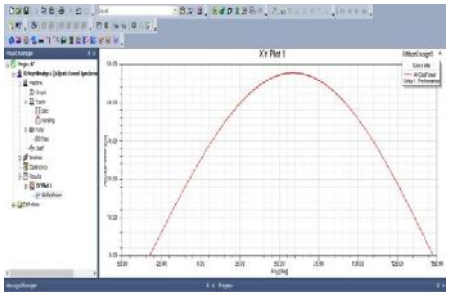

Fig 5.5 Airgap power

Fig 5.6 shows the speed curve of PMSM

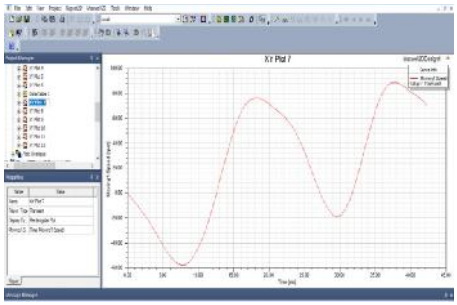

Fig 5.6 speed curve

V. CONCLUSION

In this paper, an inner rotor permanent magnet synchronous motor with 5hp is designed for electric vehicle applications. The design procedure and analysis are achieved by using Ansys Maxwell. The proposed PMSM adopting new rotor geometry, which posse's excessive power density, excessive efficiency, smooths torque, and handy speed regulation is tailor-made for the electric vehicle.Even though, in precept, the proposed PMSM can be effortlessly scaled upward to fit the needs of a regular on-road electric vehicle with a motor of order 3.5kW, the low-priced advantage may be faded because of the relatively high-priced magnetic fabric. The simulation effects provide the winding contemporary, torque, flux linkage, and velocity traits curves.

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